Dopamine depletion leads to pathological synchronization of distinct basal ganglia loops in the beta band

Paper: Ortone et al. (2023). Spiking-network modelling study.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

Possible network mechanisms of beta synchronization and the limitations of an isolated STN–GPe model.

How does it answer the general questions?

Finding. Interacting STN–GPe and striatal–GPe loops show beta-range resonances. In simulations, dopamine depletion strengthens the striatal loop and promotes synchronization between loops that are more independent at high dopamine levels (PDF pp. 1–4, Results).

Usable insight. The source of a shared spectral peak can depend on interactions between circuits and dopamine-dependent coupling. Network interventions should therefore be evaluated across mechanisms and operating conditions. The paper establishes a computational explanation of synchronization; it does not demonstrate that minimizing beta improves patients' motor function.

Model relevance. Six populations are included: D1/D2 medium spiny neurons, fast-spiking interneurons, two GPe populations and STN. Individual neurons have adaptation, conductance dynamics and reset events. These equations are not exactly the notebook's memoryless monotone rate model. Tables 2–3 give biophysical parameters and projection delays; they require a justified population reduction before becoming rate-model coefficients (PDF pp. 4–7).

What papers does it cite that could provide more information?

- De la Crompe et al. (2020), *The globus pallidus orchestrates abnormal network dynamics in a model of Parkinsonism*: experimental constraint (ref. 49, PDF p. 28).
- Corbit et al. (2016), *Pallidostriatal projections promote beta oscillations in a dopamine-depleted biophysical network model*: striatal-loop comparison (ref. 57, PDF pp. 28–29).